Extreme Cases: Why G2 Dominates G1 at the Query Level

Deep dive — 10 queries with largest G2–G1 gap

> *We identify the 10 queries where the centroid-based retrieval group (G2) most dramatically exceeds the query-centered group (G1). Two distinct mechanisms explain the gap. In both, the root cause is the same: queries occupy a sparse region of embedding space, while documents cluster in a structurally denser zone that the centroid of LLM-scored documents (G3) consistently reaches.*

RESULTS TABLE

#	QUERY	G1	G2	G3	GAP	R1	R2	MECHANISM	RAG
1	118	1	68	4	67	0.158	0.331	A	
2	37	1	42	31	41	0.200	0.153	B	
3	119	1	38	20	37	0.221	0.153	B	
4	190	25	56	25	31	0.292	0.223	B	FAIL
5	29	1	26	10	25	0.235	0.245	A	
6	96	1	24	19	23	0.247	0.174	B	
7	43	0	22	14	22	0.289	0.166	B	
8	85	1	23	15	22	0.076	0.081	A	
9	114	1	23	19	22	0.140	0.138	B	
10	153	2	24	33	22	0.183	0.143	B	FAIL

r1 = dist(Query, GT) • r2 = dist(Centroid, GT). G1 is 0 or 1 in 9 of 10 cases.

TWO MECHANISMS

MECHANISM A — 3 QUERIES (118, 29, 85)

The G3 centroid is pulled *away* from GT by false-positive documents, so r2 ≥ r1. Yet G2 is large because the centroid lands inside a structurally dense zone of document space. Proximity to GT is not required — presence in document space is sufficient.

`Query 118: r2 = 2.1 × r1, G2/G1 = 68×`

MECHANISM B — 7 QUERIES

G3 forms a coherent, tight cluster. Its centroid falls close to GT (r2 < r1). Even a smaller centroid circle lands inside the dense document cluster around GT and captures many more documents than G1.

`Median G3 spread std: 0.088 (vs 0.726 in Mech A)`

CONCLUSIONS

1. **G1 collapses to 0-1 documents in 9 of 10 extreme cases.** Queries sit in a structurally sparse region of embedding space; almost nothing is near the query even when the GT is nearby.

2. **The centroid does not need to be close to GT.** Query 118 has r2 more than twice r1, yet G2 = 68. What matters is that the centroid lands in document space, which is inherently denser than query space.

3. **Two paths to a large G2.** Mechanism A: centroid displaced by false positives, lands in dense document region. Mechanism B: tight G3 cluster, centroid genuinely close to GT. Both produce the same result: G2 >> G1.

4. **RAG failure does not prevent a large G2.** Queries 190 and 153 (RAG failed, GT not in G3) still reach G2 = 56 and 24. G2 reflects the coherent document cluster, not the correctness of scoring.

5. **The density asymmetry between query space and document space is the fundamental cause** of the G2 > G1 phenomenon across all 196 queries, and it is most visible at the extremes studied here.

G1: docs within dist(Q, GT) of the query. G2: docs within dist(Centroid_{G3}, GT) of the G3 centroid. G3: LLM-scored docs with score = 1.0. Script: `analyze_extreme_examples.py` — reading pre-computed output of `analyze_groups.py`.